

Akshay Trivedi (he/him)

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Professional Experience

Astera Labs, Firmware Engineer

Jun 2025—Aug 2026

San Jose, CA

Developed firmware for PCIe Gen6 switches, which play a significant role in Astera's rack-scale AI and compute platform. Worked in all three layers of PCIe - Transaction, Data Link, and Physical. Gained detailed knowledge of the PCIe 6.0 standard.

Built firmware infrastructure and management modules leveraging an in-depth understanding of the system architecture. Eliminated some complex bugs in the RTOS and interrupt-handling logic.

Astera Labs, Software Engineer Intern

May 2024—Aug 2024

Santa Clara, CA

Created a toolchain for configuring ASIC firmware, and developed infrastructure for testing pre-silicon ASIC firmware on an emulation platform. Used Jenkins, pytest, I2C.

Gained some familiarity with the PCIe 6.0 standard.

TabaPay, Software Engineer Intern

May 2023—Aug 2023

Mountain View, CA

Designed and implemented an internal database for TabaPay employees to quickly view their clients' API configuration state. Used Golang and MySQL.

Learned how the card acceptor industry works and gained valuable experience working in a fast-paced and dynamic startup environment.

Education

Carnegie Mellon University, MS in CS

Aug 2026—Dec 2027

Pittsburgh, PA

ML (10601), Dist. Systems (15640), Algorithms (15651).

University of Maryland, BS in CS and Math

Aug 2022—May 2025

College Park, MD

Networks (CMSC711, CMSC417), OS (CMSC412), Computer Arch. (CMSC411), Cryptography (CMSC456), Parallel Comp. (CMSC416), Algorithms (CMSC451), Data Structures (CMSC420). Num. Analysis I (AMSC466), Lin. Alg. (MATH405), Combinatorics & Graph Thy (MATH475), Axiomatic Set Thy (MATH446), Abstract Alg. (MATH403), Adv. Calc. I (MATH410), Prob. Thy (STAT410).

GPA: 4.000

About Me

I am a systems programmer. I believe that planning, testing, and benchmarking are crucial steps in building software. I design solutions that use the available hardware resources efficiently. I enjoy tackling problems under constraints.

I am a fast learner; I'm good at understanding documentation and requirements to determine the best solution to a problem.

I also know that I cannot solve the hardest problems on my own, and I am great at working with others to get things done. I am a team player; I ask for and provide help in a positive and constructive manner.

Skills

Programming Languages: Zig, Python, Rust, C/C++, Golang, Java.

Systems programming (firmware and software): RTOS, assembly.

PCIe (up to Gen 6).

Graphics programming: OpenGL, Vulkan, and WebGPU.

Awards

Dean's List; Fall 2022 to Spring 2025 - University of Maryland

[Grant Family Outstanding Achievement Undergraduate Student Award in Computer Science and Mathematics](#); May 2025 - University of Maryland